

SOLUTIONS

1.

TL	L	T-Th	Th	H	T	O
3	2	5	4	7	1	0

Place value of 5 = $(5 \times 10000) = 50000$.

2. The face value of 8 in the given numeral is 8.
3. Sum of the place values of 3 = $(3000 + 30) = 3030$.
4. Difference between the place values of 7 and 3 in given numeral = $(7000 - 30) = 6970$.
5. Difference between the local value and the face value of 7 in the given numeral = $(70000 - 7) = 69993$.
6. Required sum = $(99999 + 10000) = 109999$.
7. Required difference = $(10000 - 999) = 9001$.
8. Required number = 30005.
9. Required number = 2047.
10. All natural numbers and 0 are called the whole numbers.
11. Clearly, there exists a smallest natural number, namely 1. So statement (1) is true.
Natural numbers are counting numbers and the counting process never ends. So, the largest nature number is not known. Thus, statement (2) is false.
There is no natural number between two consecutive natural numbers. So, statement (3) is false.
12. Clearly, every rational number is also a real number.
13. Clearly, π is an irrational number.
14. Since $\sqrt{2}$ is a non-terminating and non-repeating decimal, so it is an irrational number.
15. $\sqrt{3}$ is an infinite non-recurring decimal.
16. We can write $9 = (1 + 8)$; $9 = (2 + 7)$;
 $9 = (3 + 6)$, $9 = (4 + 5)$.
Thus, it can be done in 4 ways.
17. We may have (64 and 1), (32 and 2), (16 and 4) and (8 and 8).
In any case, the sum is not 35.
18. We may take the least values of y and z as $y = 1$ and $z = 1$.
So, the maximum value of x is 7.
19. We know that: $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6} n(n+1)(2n+1)$.
 $\therefore (1^2 + 2^2 + 3^2 + \dots + 9^2) = \left(\frac{1}{6} \times 9 \times 10 \times 19 \right) = 285$.
20. $(n + 7) = 88 \Rightarrow n = (88 - 7) = 81$, which is false as $20 < n < 80$.
So, the required number is 88.
21. We have, $M = 1000$; $D = 500$; $C = 100$ and $L = 50$.
 $\therefore M > D > C > L$ is the correct sequence.
22. These numbers are 24, 22, 20, 18, 16, 14, 12, 10, 8, 6, 4, 2. 8th number from the bottom is 16.
23. The given series is such that the sum of first hundred terms is zero, and 101st term is 2. So, the sum of 101 terms is 2.
24. Given series is 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4,
 $\therefore$ 96th term is 4. So, $T_{97} = 1$ and $T_{98} = 2$. Hence, the 98th term is 2.
25. Let the hundred's, ten's and one's digits be x , y and z respectively.
Then, the given number is $100x + 10y + z$.
26. Let the thousand's, hundred's, ten's and one's digits be x , y , z , w respectively.

Then, the number is $1000x + 100y + 10z + w = 10^3x + 10^2y + 10z + w$.

27. We know that the sum of two odd numbers is even.
 $\therefore (n \text{ is odd, } p \text{ is odd}) \Rightarrow (n + p) \text{ is even.}$
28. n^3 is odd $\Rightarrow n$ is odd and n^2 is odd.
 $\therefore$ I and II are true.
29. $(n - 1)$ is odd $\Rightarrow (n - 1) - 2$ and $(n - 1) + 2$ are odd $\Rightarrow (n - 3)$ and $(n + 1)$ are odd.
30. The product of two odd numbers is always odd.
31. x is odd $\Rightarrow x^2$ is odd $\Rightarrow 5x^2$ is odd $\Rightarrow (5x^2 + 2)$ is odd.
32. $ab = 0 \Rightarrow a = 0$ or $b = 0$ or both are zero.
33. $A < B < C < D \Rightarrow D > B > A$
Also, $A < B < C < D$ and $D > B > E \Rightarrow E < B < C < D \Rightarrow E < C < D$ and $E < B < C$.
Clearly, we cannot arrange A, E, C in an increasing or decreasing order.
34. When m is even, then $m(n + 'o')(p - q)$ is even.
35. $n < 0 \Rightarrow 2n < 0$, $-n > 0$ and $n^2 = (-n)^2 > 0$.
Thus, out of the numbers 0, $-n$, $2n$ and n^2 we find that $2n$ is the least.
36. It is given that $x - y = 8$.
I. We may have $x = 5$ and $y = -3$.
So, it is not necessary that both x and y are positive.
II. If x is positive, then it is not necessary that y is positive, as $x = 5 \Rightarrow y = -3$.
III. If $x < 0$, then $y = x - 8$ which is clearly less than 0.
So, if x is negative, then y must be negative.
37. If $x < 0$ and $y < 0$, then clearly $xy > 0$.
So, whenever x and y are negative, then xy is positive.
Note that $(x < 0, y < 0)$ does not imply that $(x + y)$ is positive.
e.g. If $x = -2$ and $y = -3$, then $(x + y) = -5 < 0$.
Note that $(x < 0, y < 0)$ does not imply that $(x - y) > 0$.
e.g. If $x = -5$ and $y = -2$, then $x - y = -5 - (-2) = -5 + 2 = -3 < 0$.
38. Let $n = 1 + x = 1 + m(m + 1)(m + 2)(m + 3)$, where m is a positive integer.
Then, clearly two of $m, (m + 1), (m + 2), (m + 3)$ are even and so their product is even. Thus, x is even and hence $n = 1 + x$ is odd.
Also, $n = 1 + m(m + 3)(m + 1)(m + 2) = 1 + (m^2 + 3m)(m^2 + 3m + 2)$
 $\Rightarrow n = 1 + y(y + 2)$, where $m^2 + 3m = y$
 $\Rightarrow n = 1 + y^2 + 2y = (1 + y)^2$, which is a perfect square.
Hence, I and III are true.
39. $1 \leq x \leq 2 \Rightarrow 1 \leq \frac{2}{5}y + 3 \leq 2$
 $\Rightarrow (1 - 3) \leq \frac{2}{5}y \leq (2 - 3) \Rightarrow \frac{5}{2}(1 - 3) \leq y \leq \frac{5}{2}(2 - 3)$
 $\Rightarrow -5 \leq y \leq \frac{-5}{2}$.
Hence, y increases from -5 to $\frac{-5}{2}$.
40. (a) Let $x = 0$ and $y = \sqrt{2}$. Then, x is rational and y is irrational.
 $\therefore x + y = 0 + \sqrt{2} = \sqrt{2}$ which is irrational.
Thus, $x + y$ is not rational.